Kundai Farai Sachikonye

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github.com/fullscreen-triangle

Computational biologist: multi-omics, systems biology, and mass spectrometry, with machine learning and strong software engineering — from method to deployed tool.

Profile

Computational biologist working across genomics / transcriptomics, proteomics / metabolomics, and systems-biology modelling. I combine machine learning with rigorous software engineering, and validate against authoritative reference data.

Technical Skills

Omics	Genomics, transcriptomics, proteomics, metabolomics; multi-omics integration
Systems biology	Cellular and circuit modelling; pathway and network analysis
ML & statistics	PyTorch, scikit-learn; statistical modelling; representation learning
Software & data	Python (primary), R, Rust, SQL; HPC (Ray, Dask, SLURM); reproducible pipelines; database design (lipidomics)
Validation	NIST, IEDB, 1000 Genomes / gnomAD / UK Biobank

Experience

2021–present **Independent Computational Biology Researcher**, *Bitspark GmbH*
Multi-omics, systems-biology, and mass-spectrometry tools; deployed, documented, reference-validated.

2019–2021 **Computational Lipidomics**, *Technische Universität München*, Munich
High-throughput mass spectrometry and multi-omics analysis at HPC scale. Taught at Master's level.

2017–2018 **Clinical Glycomics & Proteomics**, *Universitätsklinikum Hamburg-Eppendorf*, Hamburg
LC-MS/MS and MALDI-TOF analysis for clinical samples.

2013 **Cancer Biomarker Discovery**, *University Medical Center Groningen*, Groningen
UPLC-MS/MS shotgun proteomics on tumour samples.

Selected Projects

gospel	Multi-omics integration (genotype + transcriptomics + metagenomics); validated against 1000 Genomes, gnomAD, UK Biobank. github.com/fullscreen-triangle/gospel
systems-biology-shaders	Computational cell / systems-biology modelling (GPU pipeline). systems-biology-shaders.vercel.app
syndrome	Computational oncology — neoantigen / MHC prediction (AUC 0.81, IEDB). syndrome-seven.vercel.app

Crown Prince Computational pharmacology (PK / PD / ADME); NIST-validated. crown-prince-four-courners.vercel.app

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*

2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*

2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*

Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

English Native / Fluent (C2)

German Conversational *resident in Germany since 2011*